



K A R M A

KARMA AUTOMOTIVE
POWERED BY KARMA
ELECTRIC
E-450



MODEL YEAR(S) 2014 +



INFORMATION FOR FIRST AND SECOND RESPONDERS EMERGENCY RESPONSE GUIDE





KARMA

FIRST RESPONDER GUIDE

Karma Automotive, Powered by Karma E-450

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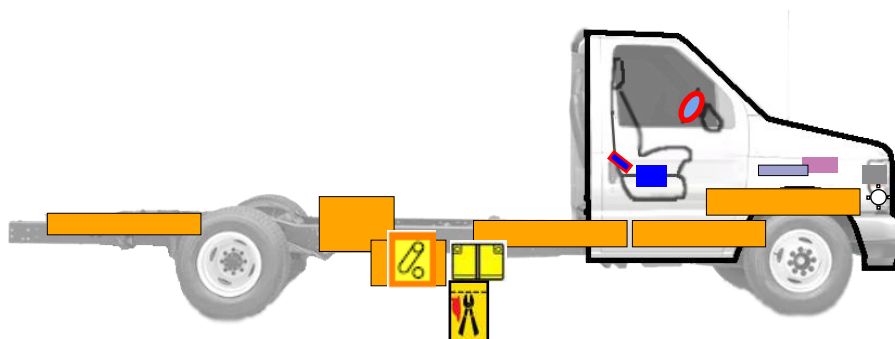
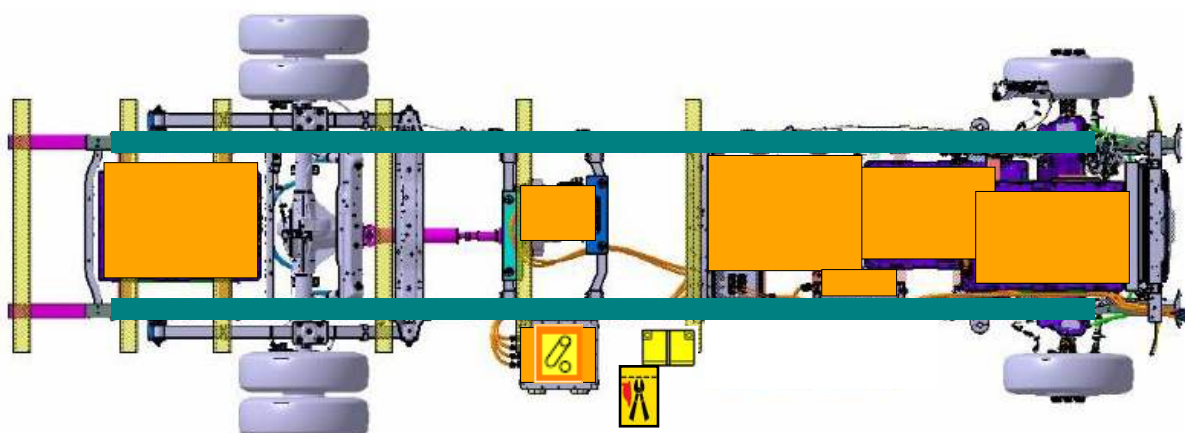
KARMA AUTOMOTIVE
POWERED BY KARMA FORD E-450 2014-2019,
AND 2021-2024

1. Identification / recognition



WARNING!

High Voltage system with DC voltage up to 400V. Do not touch high voltage components. These items can be identified by the orange cables connected to them



	High Voltage component		PTC heater		SRS control module
	12 Volt Battery		HV Charging Connector		Seatbelt Tensioner
	Low voltage disable cut points		A/C		Airbag
	High voltage disconnect points		Fusebox		Structural reinforcements



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Identification / recognition



WARNING

Approach a damaged vehicle or vehicle in the need of recovery with caution. Lack of any sounds (engine or other) does not mean the vehicle is powered off. Silent movement or a restart can occur until the vehicle is completely shut down and powered off. Wear appropriate PPE.

A Powered by Karma vehicle can be identified by the visible charge port on the front grille.

This is a rear drive vehicle with a traction motor and inverter unit located centrally mounted in the chassis. Some variants will exist for this powertrain on a E-450 cut away chassis, all safety warnings will be marked appropriately.





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2. Immobilization / stabilization / lifting

Immobilization

Chock rear tires/wheels to keep the Karma Vehicle from rolling backward or forward. Always chock the wheels.



Power off Vehicle

Pressing the power mode button will power off the vehicle if all systems are normal. This is located the the right of the steering wheel on the dash board, it is illuminated when the vehicle is powered on.

Put Vehicle in Park

Karma Vehicle is powered on silently, there is an audible passenger sound when the vehicle is in motion however, never assume it is powered off when stationary. Pressing the accelerator pedal while a Karma Vehicle is in Drive or Reverse may result in very rapid acceleration. Press the button located on Drive mode selector located on the dash. Once in Park, a "P" will be indicated along the lower edge of the center gauge cluster, and the electronic parking brake should engage to immobilize the rear wheels. if powered on the driver information panel will indicate park is applied with a "P" in the center display as shown below.



PRND indicator, illuminates the selected drive mode.



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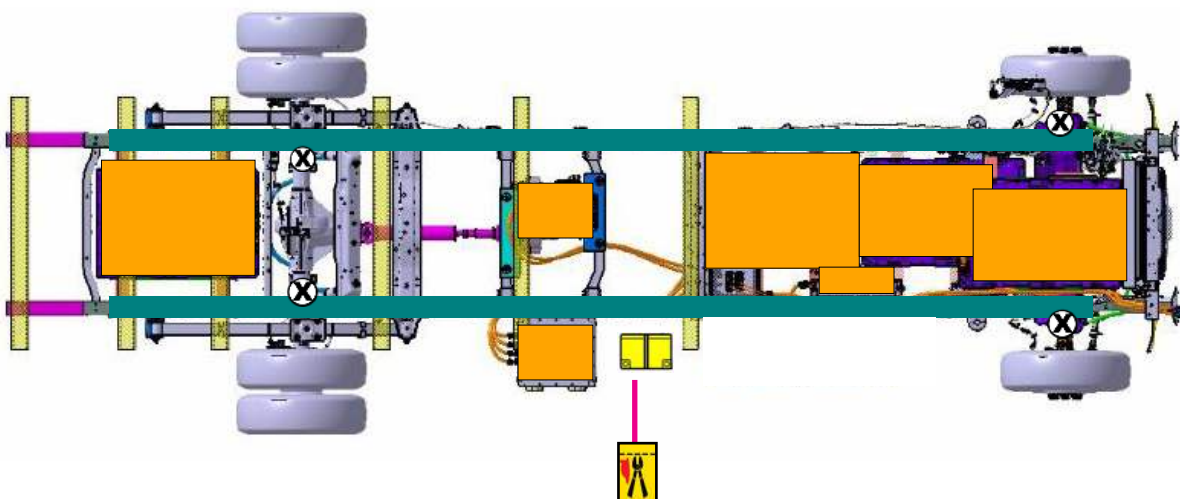
Immobilization / stabilization / lifting



WARNING

The vehicle should only be lifted or manipulated if the first responders are trained and equipped at the technician level per the applicable country's national fire training requirements and are familiar with the vehicle's lifting points. Use caution to ensure you avoid contact with the high-voltage battery or other high-voltage components while lifting or manipulating the vehicle. Avoid contact with the high voltage battery.

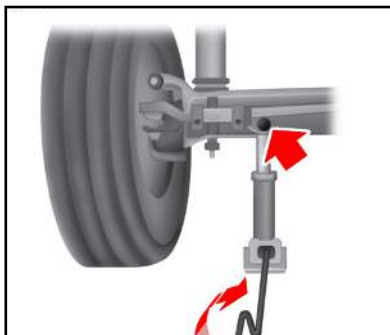
This is a body on frame design, do not use the fitted cab or body as a lift point.



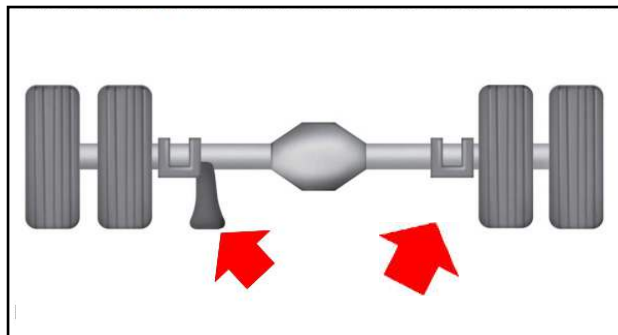
High voltage
High strength zone

X Lift point

Place the jack under the front axle pin



Do not place jack under steering linkage



Place jack or lifting as indicated.



3. Disable direct hazards / safety regulations

Ensure Passengers are out of the vehicle prior to cutting low voltage power!

Emergency Low Voltage Disable Locations

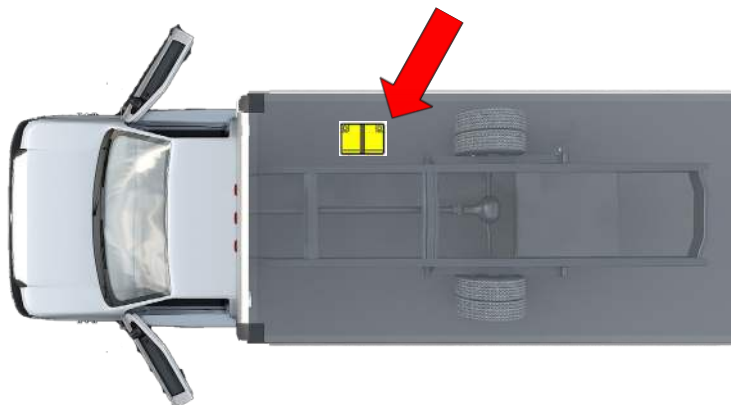
If access to the MSD cannot be gained, along side is the 12 volt battery location, on the frame rail. The harness with a clearly visible yellow label, with a scissor illustration. Cut one or both of the cables at the appointed line, this is a low voltage safe system.

Unplugging/Cutting the first responder loop removes low-voltage power going to the high-voltage contactors inside of the high-voltage battery pack. Disconnecting the first responder loop does not disable the low-voltage battery system. The vehicle's low voltage (12-volt battery) is located on the passenger entry (curb side) side, on the frame in a carrier aft of the front wheel. The battery must be disconnected or low voltage cables cut in order to disable the low-voltage battery system, which powers the Supplemental Restraint System ("SRS") control unit.

NOTE: Once the first responder loop has been cut, all Karma-retrofitted systems, including the Electronic Parking Brake System and High Voltage Battery will be inoperable. If 12-volt Battery cables such as the main battery ground have been disconnected or cut, all Karma-retrofitted systems as noted, and Ford-factory systems such as power doors, windows, seats and steering column will be inoperable. Manual door release levers are visible and vary per vehicle based on body up-fitter specification.



The cut locations will have the cut tag on the appropriate harness on the red cables from the battery.





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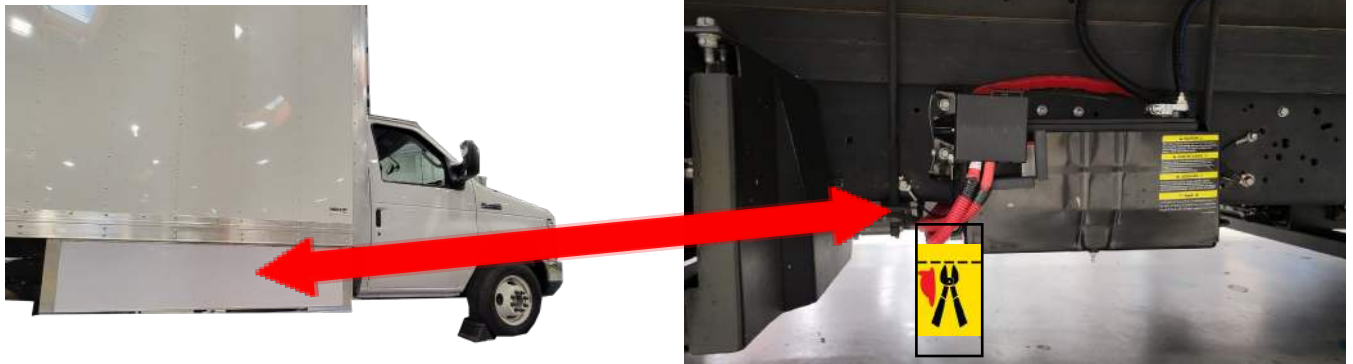
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Disable direct hazards / safety regulations

Low Voltage Battery

The hybrid has a 12-Volt maintenance Free Battery which is located on the passenger frame rail. The Absorbent Glass-Mat ("AGM") battery stores 12-Volt power for the vehicle's low voltage system. If the low voltage cable is not cut, disconnecting the cables at the battery is a service method to disconnect the low voltage system.



Curb side frame mounted 12-Volt battery and
low voltage (positive cable) disable location

The hood can be opened by the manual hood release cable located on the left side knee bolster area on the driver side in the vehicle. This will provide access the fuse box and HV A/C compressor.



WARNING

Not every High-Voltage component is labeled. Always wear appropriate PPE.
Do not attempt to open the High-Voltage Battery.



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Disable direct hazards / safety regulations

Even if the 12 Volt loop has been cut, always treat the high-voltage pack and the high-voltage components as if these are live. Treat every battery pack and orange cable as if there is high voltage in them. Never cut an orange high-voltage cable or cut into the battery pack.

There is no way to perform an instant discharge of the energy inside of the battery pack when a vehicle is in an accident. There is stored energy in the high-voltage battery. Caution must be used to not damage the battery pack, in the case of vehicle extrication operations.

Vehicle Charging

In the case of an emergency incident that involves collision, electrical failure, or fire, while charging additional steps should be taken to isolate the charging station. If the charging station is equipped with an e-stop, it should be applied. If an e-stop is not present or accessible, property management should be accessed to locate the main service disconnect and lockout/tagout breaker. After it has been located, immediately disconnect the charger handle at the vehicle by pressing the release button on the charging cable handle.



Charger release button on handle



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Disable direct hazards / safety regulations

Disable High Voltage Electrical System

The Manual Service Disconnect (“MSD”) is on the passenger entrance side of the vehicle. Some versions may have a ‘vanity’ panel shrouding the MSD and 12Volt battery location. Access will be behind this.



MSDs installed



MSDs removed in this image

following page illustrates removal of MSD



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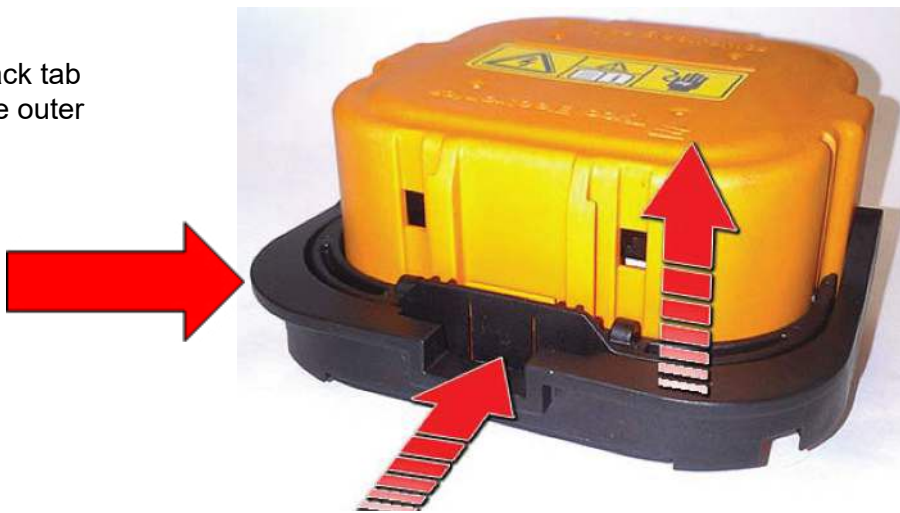
Disable direct hazards / safety regulations



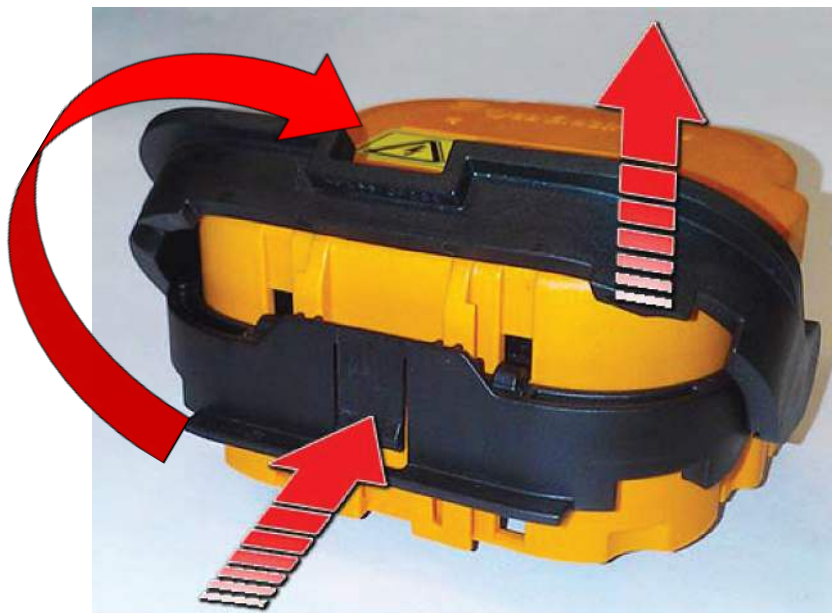
WARNING

You must wait 5 minutes after the MSD cap has been removed. Although the battery is now isolated from the system, the harness may need 5 minutes to discharge the harness and High Voltage Capacitors internal to the High Voltage components.

Press the black tab and move the outer handle.



Lift outer handle rotate 45 degrees (as shown), where it will stop. Press the black tab on the next part of the handle. Pull up on the handle to disengage the Manual Service Disconnect for the HV battery S-box location.





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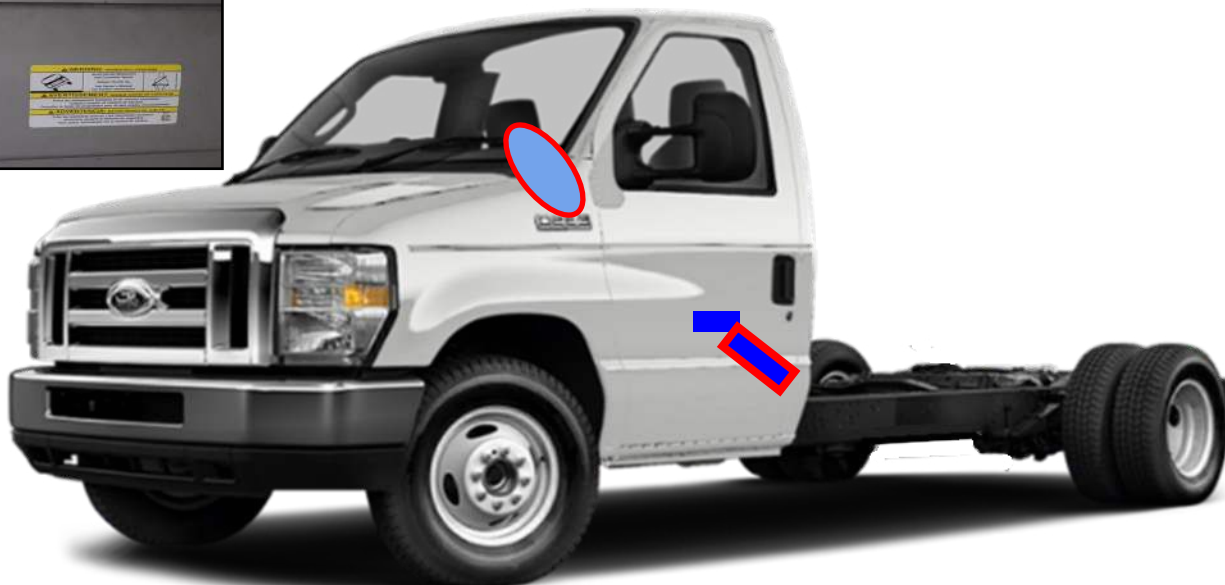
AIRBAGS



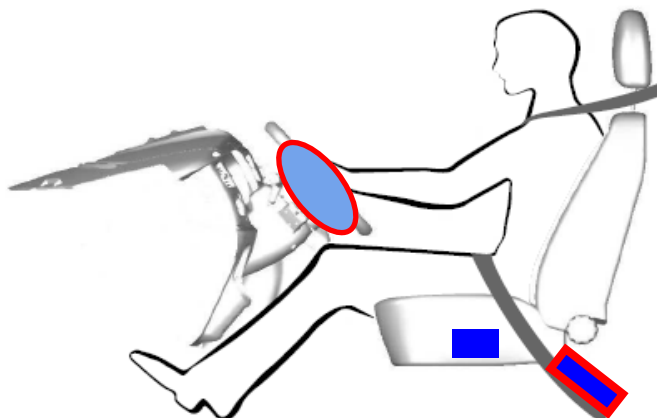
WARNING

The supplementary restraint system (“SRS”) unit has an internal energy reserve, which enables it to remain powered for up to five minutes after Low-Voltage battery power has been disconnected. The SRS unit will remain powered from the vehicle after it deploys any airbag or per-tensioner. Do not touch the SRS unit while it is powered. The SRS control unit has a backup power reserve with a discharge time of approximately ten seconds. Do not touch the SRS control unit within ten seconds of an airbag or per-tensioner deployment.

Airbags are located in the approximate areas shown. Airbag warning information is printed on the sun visors. The Low-Voltage Battery negative cables (2) should be cut (or if possible removed from the battery terminal) in order to open the low voltage circuit that provides power to the airbags.



1. Driver airbag 
2. SRS control module under driver seat 
3. Seatbelt Tensioner 



WARNING

NO CUT ZONES. Never cut or crush inflation cylinders. Cutting or compressing cylinders can cause catastrophic failure and lead to serious injury or death.



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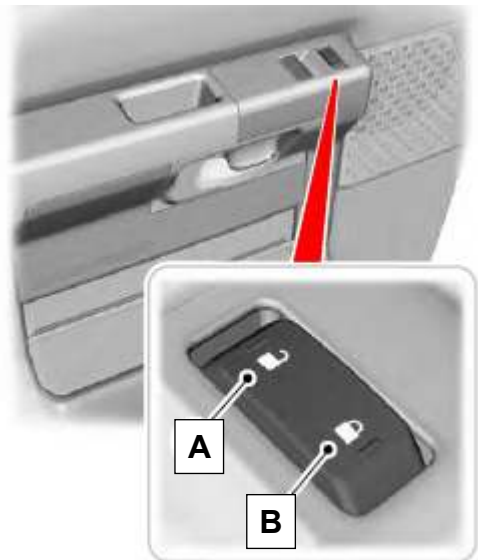
4. Access to occupants

Access to the vehicle is done using the door handles. To unlock use the Key. on the driver side. Entry through the passenger side may vary on configuration, and the release is found inside the passenger compartment

Lift outside door handle, and pull door open



From the interior use the door unlock and release, press button (A to unlock B to lock), or pull the handle and push the door open.



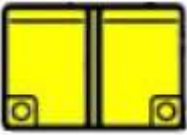


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5. Stored energy / liquids / gases / solids

	  	12 Volts Sealed AGM
	     	Lithium Ion 400 Volts
	 	HFC134A, A/C refrigerant (up to 2.8 kg with two compressors)
High Voltage System Coolant	 	G48 (HOAT)Coolant Blue (22 liters)
Electric drive Components Coolant	 	G48 Engine (HOAT) Coolant Blue (15 liters)



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High Voltage Electrical System Information



WARNING



ELECTRIC AND HYBRID VEHICLES DAMAGED BY A CRASH MAY HAVE COMPROMISED HIGH VOLTAGE SAFETY SYSTEMS AND PRESENT A POTENTIAL HIGH VOLTAGE ELECTRICAL SHOCK HAZARD. EXERCISE CAUTION ALWAYS WEAR APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT (PPE) INCLUDING HIGH VOLTAGE SAFETY GLOVES AND BOOTS. REMOVE ALL METALLIC JEWELRY, INCLUDING WATCHES AND RINGS. ISOLATE THE HIGH VOLTAGE SYSTEM AS DIRECTED BY THIS EMERGENCY RESPONSE GUIDE FOR THE VEHICLE. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

The following features have been incorporated into Powered by Karma Vehicles to allow for shut-off of the High Voltage electrical systems, or automatic shut-off.

Manual disconnect with high voltage fuse — In the event of a High Voltage short circuit, the High Voltage fuse opens, isolating the High Voltage battery from the circuit.

Power in the OFF Position — Any time the power is in the off position the High Voltage system is isolated to the battery pack, this includes if the charge cable is not connected to the vehicle charge port.

PLEASE NOTE not every crash will cause the vehicle to shutoff or High Voltage system deactivation.



WARNING



FIRES IN CRASH-DAMAGED ELECTRIC VEHICLES MAY EMIT TOXIC OR COMBUSTIBLE GASSES. SOME AMOUNTS OF EYE, SKIN OR LUNG IRRITANTS MAY BE PRESENT. WEAR PERSONAL PROTECTIVE EQUIPMENT AND SELF-CONTAINED BREATHING APPARATUS WHEN WORKING IN CLOSE PROXIMITY OR IN A CONFINED AREA (e.g. A TUNNEL OR GARAGE). VENTILATE THE VEHICLE INTERIOR BY OPENING VEHICLE DOORS OR WINDOWS. VENTILATE THE WORKING AREA. FAILURE TO FOLLOW THIS INSTRUCTION MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

High Voltage warning labels will be present on or around high voltage components





6. In case of fire



USE WATER TO FIGHT A HIGH VOLTAGE BATTERY FIRE.

If the battery catches fire, or is generating heat or gases, or if it is exposed to high heat, use large amounts of water to cool the battery. It can take between approximately 3,000-8,000 gallons (11,356-30,283 liters) of water, applied directly to the battery, to fully extinguish and cool down a battery fire; always establish or request additional water supply early. If water is not immediately available, use CO₂, dry chemicals, or another typical fire-extinguishing agent to fight the fire until water is available.

Electricity is stored in a high-voltage battery pack.
Use of Class ABC powder-type extinguisher to contain and smother the flames.

Class A	TYPICAL FIRE – Wood, Paper, Cloth, Rubber, and most Plastics
Class B	CHEMICAL FIRE – Flammable Liquids, Gasoline, Kerosene, Oil, Grease, Gases, and Solvents
Class C	ELECTRICAL FIRE – Live Electrical Equipment and Wiring

Apply water directly to the battery.

If safety permits, tilt or lift the vehicle for more direct access to the battery packs which are mounted between the frame rails.

Never open the battery for the purpose of cooling it.

Water may be applied onto the pack from a safe distance **ONLY** if a natural opening or puncture (such as a vent or opening from a collision) already exists.



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In case of fire



The use of a Thermal Imagery Camera or Infrared (TIC or IR) is recommended to monitor battery temperatures during the cooling process. Continue to use water until the battery has reached ambient temperatures or below, indicated by the thermal imagery camera. When utilizing a thermal imaging camera, allow time, once water application has stopped, for heat within the battery to transfer to the battery enclosure.

Extinguish small fires that do not involve the high voltage battery using standard vehicle firefighting procedures.



During fire extinguishing, do not make contact with any high voltage components. Always use insulated tools for fire extinguishing.

Heat and flames can compromise airbag inflators, stored gas inflation cylinders, gas struts, and other components which can result in unexpected excessive heat, which can cause inflation cylinder explosion.



Battery fires can take up to 24 hours to fully cool. After suppression and smoke has visibly subsided, a thermal imaging camera can be used to actively measure the temperature of the high voltage battery and monitor the trend of heating or cooling.



There must be no fire, smoke, audible popping or hissing, or heating present in the high voltage battery for at least 45 minutes before the vehicle can be released to second responders. The battery must be completely cooled before releasing the vehicle to second responders or otherwise leaving the incident. Second responders must be aware that there is a risk of battery re-ignition. Second responders should be advised to position the vehicle to drain excess water out of the vehicle by tilting or repositioning it. This operation can assist in mitigating possible re-ignition. During all firefighting activities, consider the vehicle energized. Always wear full PPE including a Self-Contained Breathing Apparatus ("SCBA").



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7. In case of submersion

Vehicles that have been exposed to large amounts of water or submerged should receive a risk analysis assessment. Contact the local Fire authorities for details.

Submerged vehicles are highly susceptible to internal short circuiting, resulting in thermal runaway. This may occur immediately upon removal or may be significantly delayed. Submerged vehicles should be staged in a safe area, a minimum of 50 ft radius, upon removal and monitored for thermal and physical indications of thermal event for 45 minutes prior to transport.



Treat a submerged Powered by Karma E-450 like any other submerged vehicle.

However, wear appropriate PPE when handling any submerged vehicle.

Remove the vehicle from the water and continue with normal high-voltage disabling.

Vehicles that have been submerged in water should be handled with greater caution due to the potential risk of a high-voltage electrical battery fire. First responders should be prepared to respond to a potential fire risk. Raise one end of the vehicle to allow water to drain out of the vehicle and the high-voltage battery pack. If the vehicle's structural integrity has been compromised, elevate the undamaged end of the vehicle to allow for proper drainage. After the vehicle is removed from the water, continue normal disabling procedures as above for 12-volt and high voltage.



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8. Towing / transportation / storage



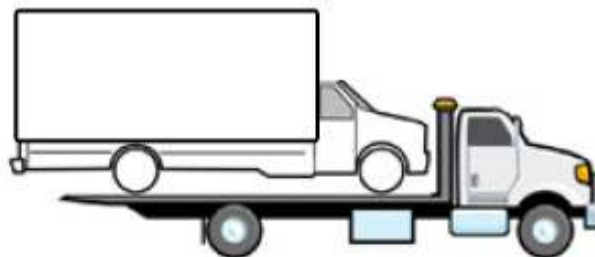
CAUTION

Pulling the vehicle with the wheels on the ground, or on a suspended lift, may cause serious damage to the vehicle as well as generating high voltages in the vehicle's electrical components.

The only approved method of recovering or transporting your vehicle is using a flatbed trailer, or transporter



Flatbed recovery





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9. Important additional information



WARNING



DEPOWERING THE HIGH VOLTAGE SYSTEM DOES NOT DISSIPATE VOLTAGE INSIDE THE BATTERY, THE BATTERY PACK REMAINS LIVE AND DANGEROUS. CONTACT WITH THE HIGH VOLTAGE BATTERY PACK INTERNALS MAY RESULT IN SERIOUS PERSONAL INJURY OR DEATH

Observe the following precautions when working on or around High Voltage batteries:
Do not cut the High Voltage battery case. Do not penetrate the batteries or case in any way.

the High Voltage battery pack is located in the central tunnel under the vehicle.

The total voltage of the vehicle HV battery pack may be up to approximately 400 volts DC.

The battery case is water resistant.

The battery pack is composed of large modules constructed of pouch cells containing gel-type liquid electrolyte. The electrolyte will not leak from the battery under most conditions. However, if the battery is crushed, it is possible for a small amount of electrolyte to leak.

If possible, isolate and avoid contact with any electric vehicle components. If contact with the High Voltage system cannot be avoided, Personal Protective Equipment (PPE) such as a splash shield or safety goggles, gloves (butyl), an apron or overcoat, and rubber boots are required when handling damaged batteries. Exposure to electrolyte could cause skin and/or eye irritation/burns. If exposed, rinse with large amounts of water for 10-15 minutes.

The High-Voltage battery will be fitted with the following label





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9. Important additional information

A Damaged vehicle or high voltage battery packs should be stored outside at a safe distance from other vehicles or structures.



WARNING

STORAGE OF DAMAGED A HIGH VOLTAGE BATTERY PACK OR A DAMAGED VEHICLE SHOULD BE KEPT AT A DISTANCE OF MINIMUM OF 15 METERS OR 50 FEET FROM OTHER STRUCTURES OR VEHICLES.



WARNING



ALWAYS ASSUME HIGH-VOLTAGE (HV) SYSTEMS TO BE ENERGIZED. DURING FIREFIGHTING ACTIVITIES, INCLUDING OVERHAUL, AVOID CONTACT WITH HV COMPONENTS. CUTTING OF HV COMPONENTS MAY CAUSE AN ARC FLASH, POTENTIALLY SEVERELY INJURING A FIREFIGHTER.



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10. Explanation of pictograms used

	General Warning		Warning, High-Voltage Electricity
	High-Voltage Battery Pack		Class B High-Voltage Propulsion System
	Cable Cutting location		Flammable
	Hazardous to Human Health		Explosive
	Acute Toxicity		Corrosives
	Gases Under Pressure		Environmental Hazard
	Infrared Thermal Camera		Extinguish with Water
	Air Conditioning Component		Warning, Low Temperature
	Lifting Point		General Caution



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